CS621-Parallel & Distributed Programming

Final Term Objective

1. In **Distributed** Control Strategy, the control function is distributed among different components in the system.

2. In **Circuit** Switching, a complete path has to be established prior to start of communication between a source and a destination.

3. A **Mesh** Topology is a network interconnection scheme in which each node is connected to every other node with a direct link.

4. The possibility to use the same software in different environment is known as **Portability**.

5. Decomposition of computation into a large number of tasks results in **fine-grained** decomposition.

6. Discrete optimization problems can be solved by using **Exploratory Decomposition** technique.

7. The problem of finding the minimum in a list is an example of decomposition of data based on **Input Data**.

8. **Recursive Decomposition** technique is more desirable for quicksort problem.

9. Sequential Consistency ensure that the result of any execution in a distributed system is equivalent to **Read and Write Operation being executed in a sequential order.**

10. What does casual consistency guarantee regarding writes in a distributed system? **All writes are seen by all processes in the same order.**

11. GPUs have gained popularity due to their ability to **provide parallel processing capabilities for certain applications**.

12. CUDA is designed for computation-intensive tasks to be executed on which hardware? **GPU**

13. GPUs accelerated libraries are commonly used in which type of application? **Data Science & Machine Learning**.

14. In message passing, how do processors exchange data in a distributed memory system? **They communicate through a communication network**.

15. What is the size of Basic MPI library in terms of the number of functions? **Basic (6 functions)**

16. In MPI, which function is commonly used to send a message from one process to another? **MPI_Send**

17. MapReduce is a software framework which supports **both parallel & distributed** computing on large data sets.

18. Which of the following is NOT true regarding Google File System (GFS)? **High Security**

19. **Descriptive Metadata** of Hadoop Distributed File System (HDFS) focuses on the individual instances of application data or the data content.

20. What is the primary function of Hadoop's HDFS? **Storing & Distributing data across a cluster of machines**.

Which component of Hadoop is responsible for storing and managing data across a cluster? **Hadoop NameNode**

21. Much of the effort associated with writing correct threaded programs is spent on **synchronizing** concurrent threads.

22. **trylock()** function is used to overcome the overhead of polling for availability of locks.

23. The length of the **longest path** in a tank dependency graph is called the critical path length.

24. In Lustre File System, which provides file service and network request handling for one or more local OSTs? **Object Storage Server(OSS)**

25. In Lustre File System, which stores file data (chunks of files)? **Object Storage Target(OST)**

26. Which is NOT true regarding POSIX I/O interface? **Many ways for describing collective access**

27. Which is NOT true regarding MP-IO? **Provides High-Level Interface**

28. Aggregates smaller read/write operation into larger operations is an example of **Buffered IO**

29. The time elapsed between the beginning and the end of its execution on a sequential computer is known as **serial runtime**

30. Speedup *S* is computed by **$S = \Theta (n / \log n)$**

31. The phenomenon when the speedup become greater than p (number of processing elements) is known as **superlinear** speedup.

32. **Cost** is the product of parallel runtime and the number of processing elements used **$p \times T_P$**

33. Using fewer than the maximum possible number of processing elements to execute a parallel algorithm is called **scaling down**

34. **Replicating** the data and moving it closer to where it is needed helps to solve the scalability problem.

Subjective

Question.No.1: Define the degree of the switch in interconnection networks.

The total number of ports on a switch is the degree of the switch. The cost of a switch grows as the square of the degree of the switch, the peripheral hardware linearly as the degree, and the packaging costs linearly as the number of pins.

Question.No.2: Bill Dally, a prominent computer scientist has highlighted power efficiency and massive parallelism as major advantages of GPUs by considering two aspects. Mention the aspects.

- Data Access Rate Capability
- Data Processing Capability

Question.No.3: Write down the names of data-centric consistency models.

- Continuous Consistency
- Consistent Ordering of Operations
- Casual Consistency
- Grouping Operations

Question.No.4: What will happen when multiple threads attempt to manipulate the same data item, if proper care is not taken to synchronize?

The results will be incoherent because they will be processing the old data rather than the new one.

Question.No.5: The I/O tools are the collection of system software and libraries have grownup to address I/O issues. List 3 I/O tools.

- At Parallel File Systems
- MPI-IO
- High level libraries

Question.No.6: Do you think, a parallel runtime of a program can be evaluated asymptotically as a function of input size only? Justify your answer in either case.

The parallel runtime of a program depends on the input size, the number of processors, the structure of the algorithm (e.g., parallelism and dependencies), and the communication parameters of the machine (e.g., latency, bandwidth, and synchronization overhead). In contrast, a sequential algorithm is evaluated by its runtime, which is typically expressed as an asymptotic function of the input size.

Question.No.7: Write down any 5 interconnection network's properties in parallel and distributed computing.

- **Diameter:** The diameter of network with n nodes is the length of the maximum shortest path between any two nodes in the network.
- **Degree of Node:** Number of connections for that node.
- **Latency:** Total time to send a message.
- **Bandwidth:** Number of bits transmitted in a unit time.
- **Bisection:** Number of connections that need to be cut to partition the network in two.

Question.No.8: The process of dividing a computation into smaller parts, some or all of which may potentially be executed in parallel, is called decomposition. You are required to list 4 key decomposition techniques.

- Recursive Decomposition
- Exploratory Decomposition
- Speculative Decomposition
- Data Decomposition

<u>Question.No.9</u>: Differentiate the statements as Independent MPI-IO and Collective MPI-IO calls.

- **Independent MPI-IO calls:**
  - ✓ MPI_File_write()
  - ✓ MPI_File_read()
- **Collective MPI-IO calls:**
  - ✓ MPI_File_write_all()
  - ✓ MPI_File_read_all()